Normative Common Ground Replication (NormCoRe): Replication-by-Translation for Studying Collective Norms in Multi-agent AI

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In the late 2010s, the fashion trend *NormCore* embraced deliberate sameness as a signal of belonging, illustrating how social norms emerge through collective coordination rather than individual preference. Today, similar forms of normative coordination increasingly arise in systems based on Multi-Agent Artificial Intelligence (MAAI), as AI-based agents deliberate, negotiate, and converge on shared decisions in fairness-sensitive domains. Yet, existing empirical approaches often treat norms as targets for alignment or replication, implicitly assuming equivalence between human subjects and AI agents and leaving collective normative dynamics insufficiently examined.

As Multi-Agent Artificial Intelligence (MAAI) systems increasingly support and automate complex tasks, they inevitably encounter decisions governed by social and ethical norms, such as fairness. Yet, empirical methods to evaluate such norms and emergent dynamics in AI agent groups remain underdeveloped. To address this gap, we propose *NormCoRe*, a novel methodological framework to systematically translate the design of human subject experiments into MAAI environments. Building on replication and MAAI research, as well as established principles from behavioral science, *NormCoRe* maps the structural layers of human subject studies onto the design of AI agent studies, allowing researchers to systematically select and document the configuration of replication studies. We demonstrate the utility of *NormCoRe* by replicating a seminal experimental study on distributive justice, in which participants negotiate fairness principles under a "veil of ignorance". Comparing AI agent groups to baselines of human groups, we uncover significant behavioral differences: while both populations favor maximizing average income subject to a floor constraint, MAAI systems exhibit substantially higher homogeneity in their consensus. Furthermore, we find that these normative judgments are sensitive to the choice of the large language model and the language of the persona description. Our work provides a rigorous pathway for investigating social norms in MAAI systems and underscores the critical need for value-sensitive design in the deployment of agentic workflows.

CCS Concepts: • **Human-centered computing → Interactive systems and tools**; • **Information systems → Decision support systems**; • **Computing methodologies → Artificial intelligence**.

Additional Key Words and Phrases: Social Norms, Ethical Norms, Experimental Studies, Replication Studies, Multi-Agent AI, Fairness, John Rawls, AI Ethics

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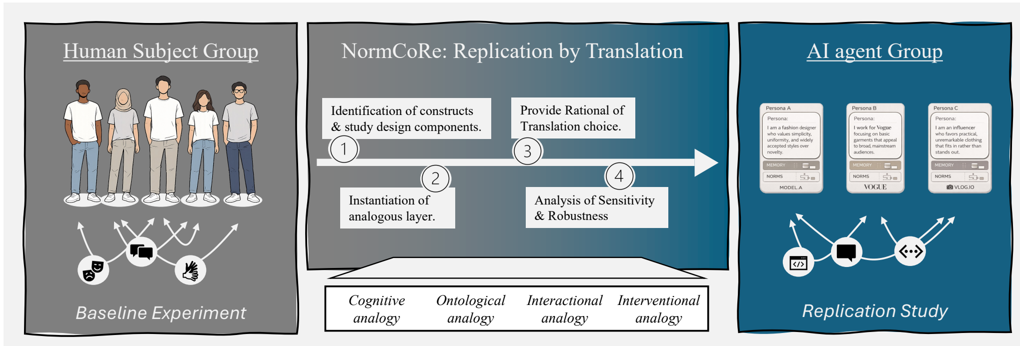

Fig. 1. From human groups to multi-agent AI: *NormCoRe* conceptualizes replication as a translation problem, mapping human subject studies to AI agent studies to study how collective normative judgments—such as fairness—emerge and differ across populations.

> *"Once upon a time people were born into communities and had to find their individuality. Today people are born individuals and have to find their communities."*
> —K-HOLE, inventors of the term *NormCore*

1 Introduction

In the late 2010s, the fashion trend *NormCore* embraced deliberate sameness: ordinary clothing as a signal of belonging rather than distinction. What appeared aesthetic was fundamentally normative; an emergent agreement about what is acceptable, appropriate—and also *fair* within a group. Such norms arise not from isolated individuals, but from collective coordination. Today, similar forms of normative coordination are increasingly produced by systems based on Multi-agent Artificial Intelligence (MAAI) illustrated in fig. 1. In MAAI [2, 23], AI-based agents now deliberate, negotiate, and converge on shared decisions. In doing so, they implicitly instantiate social and ethical norms [13]. This shift—if there is one—matters. MAAI systems are already being deployed in domains governed by fairness and other social norms, such as resource allocation [40]. If we look deeper into this field from a more traditional perspective, experimental research on social and ethical norms is deeply rooted in psychological and behavioral science [18]. But what they all have in common: They are examining *human* subjects. For example, preferences regarding wealth distribution have been extensively studied through game-theoretical group experiments [14, 16].

However, despite technological advances, empirical research has largely framed norms as targets for alignment or replication, implicitly assuming replication equivalence between human subjects and AI agents (e.g., 12, 13, 31, 34). At the same time, recent research is already proposing to replace or supplement human subjects with AI agents [4, 38], replicating existing human subject studies using AI agents [13], or attempting to align MAAI with normative principles [37]. This obscures a critical question for fairness and accountability: how collective normative judgments emerge, stabilize, and differ when decision-making is delegated to MAAI rather than human groups.

Yet, irrespective of the specific goals of such studies, the complexities arising from translating human subject studies to MAAI systems demand a sound methodological foundation that is currently lacking—particularly when social and ethical norms are the object of such studies. For example, when MAAI systems are tasked with making implicit or explicit decisions about resource allocations, the sheer number of degrees of freedom—ranging from the selection of a foundation model (e.g., Large Language Model (LLM)) to the design of task-specific workflows—impedes a thorough evaluation of design choices.

Against this backdrop, we propose Normative Common Ground Replication *NormCoRe*[1]: a novel methodological framework for translating human subject group experiments into MAAI environments, enabling researchers to *systematically* investigate social norms in AI agent studies. Building on established principles from behavioral science and state-of-the-art MAAI research, *NormCoRe* maps the analogous layers of human subject studies onto the design of AI agent studies with MAAI. This framework allows researchers to systematically select and document the configuration of AI agent studies.

We demonstrate the utility of *NormCoRe* by replicating Frohlich and Oppenheimer [16]'s influential study on fairness principles in an MAAI setting. The human subject study is particularly well suited for this purpose as it conceptualizes fairness as a group-level normative judgment reached through deliberation under uncertainty. Thus, this judgement is precisely the kind of collective process increasingly delegated to MAAI in practice. In the original experiment, Frohlich and Oppenheimer [16] operationalized John Rawls' veil of ignorance" [28] within a controlled laboratory setting. Participants were unaware of their assigned income class—simulating the "veil of ignorance"—and were tasked with reaching a consensus on a distributive justice principle that would determine their payoff in a hypothetical society. By combining individual normative deliberation, goal-based optimization, and dynamic consensus finding, this experiment serves as an ideal testbed for demonstrating how *NormCoRe* organizes complex design choices and facilitates precise reporting. Instantiating *NormCoRe* with Frohlich and Oppenheimer [16], we show that MAAI groups yield fairness judgments that diverge significantly from human groups. While both populations favor the same principle (maximizing overall income while ensuring a floor constraint for the worst-off), MAAI groups demonstrate a substantially higher concentration on this principle than human groups (29 of 33 MAAI groups, compared to 23 of 34 human groups). Furthermore, we find that the choice of the LLM and the language of the persona description have a significant influence on these fairness judgments.

This work makes two primary contributions to the value-sensitive design of MAAI systems. First, we introduce *NormCoRe* as a methodology for the systematic replication of human subject studies in MAAI settings, providing a lens through which researchers can document and understand machine social norms and compare them to the outcomes of human subject studies. Second, by employing *NormCoRe* to study fairness principles, we provide empirical evidence that social norms in MAAI can differ from human baselines and illustrate ways to analyze these differences. Our work raises critical normative questions regarding the future of agentic automation and cautions designers to make conscious, evidence-based choices in both laboratory experiments and real-world applications.

The remainder of this work is organized as follows. In Section 2, we review the challenges of replication studies in the context of MAAI. Section 3 introduces the *NormCoRe* method and its four translation layers. In **??**, we instantiate the method by detailing the original study [16], the translation methodology, and our experimental results. Section 5 discusses these findings and their limitations in relation to existing literature, and Section 6 concludes with directions for future research.

2 Background

Intro text...

[1]which, not coincidentally, bears the same name as the fashion phenomenon described above.

2.1 Importance of Replicability for the Scientific Method

Replicability constitutes a fundamental pillar of the scientific method. In principle, the transparent documentation of research methodologies enables independent verification and replication of empirical findings by the broader academic community. In that sense, replication can serve two functions: authentication of original findings and boundary testing to understand generalizability [41]. Consequently, two replication approaches are distinguished: *direct replication*, which repeats an experiment with minimal to no changes to authenticate original findings, and *conceptual replication*, which tests the same hypothesis with different methods, stimuli, or populations, aiming to probe whether findings generalize to new conditions [41]. Although sometimes treated as synonyms, a critical distinction exists between reproducibility, where existing data is re-analyzed with the same methods, and replicability, where experiments are repeated, resulting in new data [27].

2.2 Replicability of Human Subject Studies with LLMs

Meanwhile, a growing body of research across disciplines is conceptually replicating human experiments with LLM-based AI systems. These efforts pursue two distinct objectives: first, to compare human and artificial cognition as a lens for understanding the behavioral properties of AI systems; and second, to assess whether LLMs can serve as valid proxies for human participants in experimental research. Yeykelis et al. [39] assessed the replicability of consumer behavior research by replicating 133 results from 45 different studies published in the Journal of Marketing. Their methodology involved the use of LLM displaying AI personas programmed to match the original participant demographics. The researchers found that 76% of main effects and 68% of interaction effects could be reproduced. In Psychology and Management Science Cui et al. [13] replicated 156 experiments with LLMs from publications of the respective disciplines from the last 10 years. They found that in 73% and 81% of the experiments, the main effects were replicated, with success rates ranging between 46% and 63% for interaction effects. In behavioral economics Leng [22] replicate canonical prospect theory, framing, and mental accounting experiments from Kahneman and Tversky [19], Kahneman and Tversky [20], Thaler [32], Thaler and Johnson [33], and Shafir and Thaler [30]. They find that LLMs partially reproduce human mental accounting but are substantially more rational, exhibiting weak framing effects and limited transaction utility. Crucially, these behaviors vary systematically by language, with Spanish and French LLMs showing more human-like loss aversion than English and Chinese LLMs.

In the field of machine ethics, Takemoto [31] replicate the Moral Machine Experiment of Awad et al. [3], where human participants were tasked to “solve” several trolley-problem-style dilemmas. When applied to LLMs, the subsequent comparison reveals that moral judgments fluctuate depending on the specific model architecture and training selected. The implicit assumption driving this stream of research is that human consensus constitutes ethical soundness. This mirrors the naturalistic fallacy, which posits that directly inferring normative “oughts” from descriptive “is” (actual human behavior) is logically invalid [26]. Irrespective of this methodological limitation, the operational demand of this research is clear: to align AI systems with the statistical moral behavior of humans. Such approaches also gain traction in the industry, as evidenced by Anthropic’s creation and use of the GlobalOpinionQA dataset, which contains 2556 questions and human answers on ethics and current events from the World Values and Pew Global Attitudes surveys [15]. Crucially, Anthropic evaluates their LLM against the normative target that the model should reflect a country’s specific distribution of opinions when prompted, implying, for instance, that a model prompted in Russian ought to adopt a distinctively “Russian” moral perspective.

2.3 Method of AI Replication

Despite increasing popularity, studies replicating human experiments with LLM-based AI systems face a fundamental methodological challenge. Each conceptual replication requires translating human experimental constructs into AI-compatible designs. This introduces degrees of freedom at different levels of the AI system, e.g., which foundation model encodes the agent's background knowledge, how personas and prompts instantiate subject-like behavior, and which orchestration protocols structure group interaction. These translation choices are rarely documented systematically, yet they condition the very outcomes being compared. Recent research demonstrates that such choices are consequential. For example, sight variations, [29] formatting changes [36], response order, framing, and framing [10] in prompt design can alter the performance of an LLM significantly. Similarly, in multi-agent settings, modifications to deliberation protocols, including speaking order [6], debate termination timing, and adversarial structure play a decisive role in determining collective outcomes [24].

Yet, the replication studies reviewed above typically treat these design choices as implementation details rather than methodological parameters. This creates an interpretability gap: when human and AI populations produce different outcomes, it cannot be determined whether the difference reflects genuine divergence or an artifact of undocumented translation choices. Addressing this gap requires treating replication across populations as an explicit translation problem—one that renders translation decisions visible at each layer of the experimental system, from foundation models to orchestration protocols. Solely then can observed differences be attributed to specific sources rather than confounded across the translation process. To make replication mainstream in all areas of science [41], this work introduces NormCoRe as a novel methodological framework for AI agent-based replication studies.

3 Normative Common Ground Replication (NormCoRe)

NormCoRe is a systematic method for replicating *human subject studies* with *AI agent studies* in order to (empirically) investigate the normative common ground between human and AI groups. NormCoRe adapts established replication logic from the social sciences to cross-population settings involving fundamentally different kinds of groups. Rather than assuming equivalence between human subjects and AI agents, NormCoRe explicitly treats replication as a translation problem to systematically map the constructs, interventions, and interactions of human-subject studies into an analogous study with AI agents, while minimizing threats to validity. The central goal of NormCoRe is not to determine whether AI agents behave "like humans," but to identify where normative (group) judgments converge or diverge, and to attribute such differences to replication choices. This enables a disentangled investigation of questions such as whether fairness judgments in MAAI systems are sensitive to model choice, persona, or memory design, and how orchestration and coordination mechanisms shape collective decision-making trajectories.

3.1 NormCoRe as Replication-by-Translation

In contrast to direct replication [35] within a single population, cross-group replication between humans and AI agents necessarily introduces conceptual degrees of freedom arising from differences in four translation layers: *cognition analogy, ontological analogy, interactional analogy* and *interventional analogy*. We distinguish two ideal-typical translation choices as methodological parameters to implicitly acknowledge underlying differences:

- ***Literal translation for direct replication***, which aims to preserve surface-level experimental features (e.g., task structure, payoff matrices, information availability).

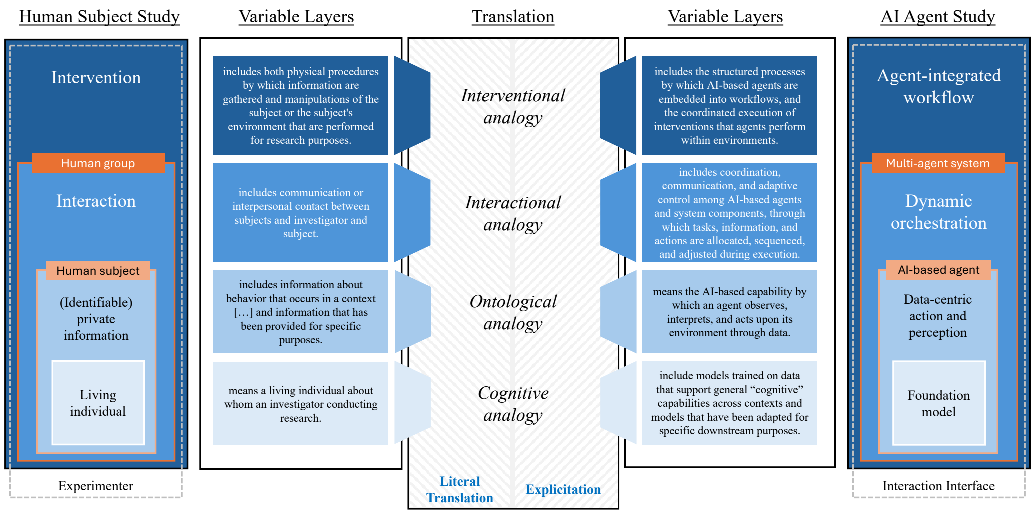

Fig. 2. The four translation layers illustrating the necessary analogies between individual layered components of human subject studies and AI agent studies. Some components may be translated "literally", e.g., when the study sequence can be fully adopted. Other components may require "explicitation", e.g., when AI agents participate in a discussion in fixed turns.

- ***Explicitation for analogous replication***, which makes implicit assumptions in human studies explicit and operationalizes them as design choices in AI agent studies (e.g., decision rules, memory structures, coordination protocols).

3.2 Layered Translation Structure in NormCoRe

NormCoRe operationalizes replication between human subject studies and AI agent studies through a layer-by-layer translation method illustrated in Figure 2. Rather than first defining human subject and AI agent studies independently, NormCoRe aligns them at the four corresponding layers of abstraction, following the nested structure of the U.S. Common Rule for human subject studies (45 CFR §46.102 [1]) in combination with the layered architecture of multi-agent AI [2]. This approach makes the translation challenge explicit at each layer, allowing normative outcomes to be interpreted relative to specific sources of variation.

Layer 1: Living Individual → Foundation Models. At the most fundamental layer, human subjects are defined as *living individuals* about whom information is obtained through intervention or interaction [1]. Normative outcomes in human studies are already shaped at this level by background knowledge, lived experience, and informational asymmetries. These factors are not themselves experimental manipulations, yet they condition all downstream perception, deliberation, and behavior. In AI agent studies, the structurally corresponding layer is the *foundation model* substrate. Foundation models encode large-scale statistical regularities derived from pretraining data and thereby constitute the epistemic and normative background against which all agent behavior unfolds [8]. While private information in human studies is generated within the experiment, foundation models represent a pre-experimental informational endowment that cannot be directly manipulated during execution but strongly conditions normative judgments.

NormCoRe treats this correspondence as a cognitive analogy. The goal is not to equate living individuals with foundation models, but to acknowledge that both define a base layer of informational constraint that shapes what kinds of cognitive heuristics or normative judgments are even expressible. Translation choices at this layer (e.g., model family, pretraining corpus, or fine-tuning approach) introduce degrees of freedom that must be documented to ensure interpretability of replication results.

Layer 2: Human Subject → AI-based Agent. Building on this informational substrate, the second layer concerns the *subject* itself. Human subjects are grounded in the notion of (identifiable) private information that is used, studied, analyzed, or generated within the research context [1]. This layer establishes the epistemic basis of the study: what is known about the subject, how that knowledge is obtained, and under which contextual constraints. This definition carries implicit assumptions about agency, perception, memory, and the capacity to form normative judgments. In AI agent studies, the corresponding layer is the agent's *data-centric perception–action capability*: the mechanisms through which an agent observes its environment, interprets inputs, and generates actions [2]. This includes prompt structures, persona specifications, memory mechanisms, and decision heuristics that instantiate agency-like behavior in computational form.

NormCoRe frames this mapping as an *ontological analogy*. Human agency is not replicated, but functionally translated into computational capacities that allow agents to participate meaningfully in normative tasks. Explicit design choices at this layer—such as whether agents possess persistent memory or adopt stable personas—directly affect normative outcomes and must therefore be treated as core elements of the replication design rather than as mere implementation details.

Layer 3: Human Group → Multi-agent System. The third layer concerns *interaction*. In human groups, interaction encompasses communication or interpersonal contact *between* subjects and investigators, as well as *among* subjects. This layer structures deliberation, persuasion, power dynamics, and social learning, all of which are central to the emergence of shared norms such as fairness [17]. In AI agent studies, the corresponding layer is *dynamic orchestration* creating a Multi-agent system. This includes message passing protocols, turn-taking rules, negotiation mechanisms, and adaptive control processes that determine how agents exchange information and influence one another over time [2]. NormCoRe treats this correspondence as an interactional analogy. Translation at this layer often requires explicitation: assumptions that are often implicit in human interaction (e.g., (equal) speaking rights, shared understanding of rules) must be made explicit as protocol constraints or coordination mechanisms in AI agent studies. Differences in normative outcomes across replications can often be traced to this layer, particularly in group-based fairness tasks [].

Layer 4: Intervention → Agent-integrated Workflow. At the highest layer, human subject studies involve *interventions*, defined as physical procedures or informational manipulations performed for research purposes [1]. Interventions include task framing, incentive structures, timing, and constraints that are intentionally varied to study causal effects. In AI agent studies, the corresponding layer is the *agent-integrated workflow* [2]. This includes the structured processes by which agents are embedded into experimental workflows, the sequencing of tasks, role assignments, and the coordinated execution of actions within an environment. This mapping constitutes an interventional analogy. NormCoRe emphasizes that such translations are not neutral: different workflow patterns can systematically shape deliberative trajectories and fairness outcomes []. Accordingly, intervention-level explicitation choices must be explicitly justified and reported.

In addition to the formally specified translation layers, the role of the experimenter should be explicitly acknowledged as an "unknown influence". In human subject studies, experimenters inevitably shape outcomes through subtle cues, framing choices, timing, and interaction styles—often unintentionally and outside the scope of formal interventions. In AI agent studies, analogous influences arise through interaction interfaces. Making this influence explicit does not eliminate it, but improves interpretability by preventing ungrounded attribution of normative differences solely to agents or models, when they may partly reflect researcher-induced artifacts. To operationalize this concern and to ensure methodological interpretability, we summarize the NormCoRe translation procedure in Box 3.2.

NormCoRe Translation Procedure

We propose a method for the interpretable replication of human subject studies with AI agent studies by establishing *disentangled, layer-specific translation analogies*, rather than assuming global equivalence. Replication success is evaluated in terms of *explanatory alignment*, not behavioral identity.

For each NormCoRe translation exercise, researchers must:

Step 1 **Identification of constructs & study design components.** Specify the theoretically relevant constructs in the original human subject study together with the concrete study design components through which these constructs are operationalized (e.g., task framing, information structure, incentive schemes, interaction rules, and experimenter involvement).

Step 2 **Instantiation of analogous layer.** Identify the MAAI component(s) that instantiate the corresponding NormCoRe layer (e.g., foundation model choice for the cognitive layer, persona prompting and memory for the ontological layer, orchestration protocols for the interactional layer).

Step 3 **Provide Rational of Translation choice.** Justify whether the layer mapping constitutes:

- *Literal translation* (for exact/direct replication), or
- *Explicitation* (for analogous replication), and specify the replication type implied by the design choice (e.g., constructive, incremental, quasirandom, or comprehensive [21]).

Step 4 **Analysis of Sensitivity & Robustness.** Analyze the sensitivity of normative outcomes to layer-specific translation choices using established statistical criteria for replication, non-replication, and partial replication evidence [9].

4 Experimental Study: Applying NormCoRe for Fairness Principles

With the methodological framework in place, we want to illustrate its application for a representative norm that has both a social and ethical dimension as well as crucial downstream impact in MAAI systems: fairness. Fairness is a norm deeply embedded in social science and philosophy, guiding perceptions, legal frameworks, economic interactions, and the design of AI systems [7]. While its meaning varies across disciplines, contexts, and cultures, several simplifying approaches have been proposed to study fairness principles in experiments with human subjects. One such experiment is Frohlich and Oppenheimer [16] who applied John Rawl's popular thought experiment of the "veil of ignorance" [28] to study preferences for four predefined fairness principles representing different schools of thought in political philosophy. The following section describes the original study from Frohlich and Oppenheimer [16] and instantiates the NormCoRe framework by translating the original study to an MAAI setting [2] to study individual judgments and group dynamics of MAAI systems in the context of fairness principles. This instantiation is meant to validate the methodological framework in a concrete application and to exemplify the method for future research.

4.1 Baseline Human Subject Study by Frohlich and Oppenheimer

Frohlich and Oppenheimer [16] conducted their experiment with 34 groups of five university students, comprising an individual and a group phase. In the individual phase, participants received an introduction to four fairness principles in the context of income distribution: (P1) maximizing the income of the worst-off individual; (P2) maximizing average (and thus total) income; (P3) maximizing average income subject to a guaranteed minimum income; and (P4) maximizing average income subject to a cap on income inequality. Participants first ranked the principles from most to least preferred and reported their confidence on a five-point Likert scale. They were then shown four alternative income distributions across five income classes with known probabilities representing a "probabilistic veil of ignorance" (5%, 10%, 50%, 25%, and 10%) and informed which distribution corresponded to each principle. After additional instruction and a comprehension test, participants ranked the principles again.

Next, participants completed four payoff-relevant practice rounds. In each round, they selected a principle, after which a corresponding distribution was implemented via a random draw assigning them to one of five income classes, essentially "lifting" the "veil of ignorance". While class probabilities were fixed, participants were unaware of their exact values. Realized payoffs, as well as counterfactual payoffs under alternative principles, were revealed and paid immediately at a 1:$10,000 conversion rate. The individual phase concluded with a third ranking and confidence assessment. In the group phase, each five-person group deliberated to reach an unanimous agreement on a single principle. Prior to the discussion, participants were informed that (i) the payoff distributions used for group payment could differ from the examples, and (ii) the group decision would determine binding payoffs at higher stakes. Unlike in the individual phase, participants did not know the specific distributions in advance. The discussion lasted at least five minutes and ended with a verbal consensus and a confirming secret-ballot vote; if unanimity was not achieved, payoffs were determined by a random draw. Finally, participants submitted a last ranking with confidence ratings.

4.2 NormCoRe Translation to AI Agents

Following NormCoRe translation procedure (cf. Box 3.2), replication is treated as a layered translation problem rather than an assumption of equivalence. Translation decisions are therefore made explicit and justified at each analogy layer of the study design.

(Step 1) Identification of constructs & study design components. The baseline experiment investigates normative preferences for distributive justice principles under a Rawlsian veil of ignorance. The focal construct is the selection and ranking of four predefined justice principles (P1–P4), measured at both the individual level (repeated rank-orderings with confidence) and the group level (unanimous consensus determining payoffs). Core design components that operationalize this construct include: (i) controlled information about income distributions and probabilities, (ii) payoff-relevant decision making via stochastic assignment to income classes, and (iii) structured group deliberation culminating in a binding collective choice. For replication validity, these elements define the non-negotiable backbone of the study. All translation choices in the AI agent study are evaluated relative to their ability to preserve this fairness decision problem while rendering it executable for LLM-based agents.

(Step 2) Instantiation of analogous layer. NormCoRe aligns the human-subject experiment with the AI-agent study through a set of layered analogies. A complete, parameter-level specification of all mappings is provided in tables 2 to 4 and **??**; here we summarize the most salient aspects.

Cognitive and ontological analogy. Human participants—university students deliberating under uncertainty—are mapped to LLM-based AI agents. In the baseline study, subjects' cognitive capacities (e.g., language comprehension, memory, reasoning ability) and ontological properties (e.g., agency, identity, persistence across interactions) are implicit and embodied. In the AI agent study, these properties must be made explicit and operationalized. Accordingly, agents are instantiated with configurable role descriptions approximating the original participant pool, managed memory with explicit character limits to support learning-in-context, and controlled linguistic and stochastic parameters (language choice and temperature). This translation preserves the functional role of the subject—forming and revising normative judgments—without asserting equivalence between human subjects and LLM-based agents.

Interactional and interventional analogies. Human group interaction and experimental procedures are translated into a turn-based orchestration protocol and an agent-integrated workflow. Free-form discussion is formalized as sequential speaking with equal opportunity, shared discussion history, and a structured consensus mechanism, while the experimenter's role and laboratory environment are translated into structured prompts, computational execution, symbolic payoffs, and explicit randomness controls. These mappings ensure that deliberation, incentives, and information flow remain comparable across populations while accommodating the technical constraints of AI agents. Full details of these mappings are documented in tables 2 to 4 and **??**.

(Step 3) Provide Rational of Translation choice. All translation choices are classified; a comprehensive classification and justification for each design decision is provided in tables 2 to 4 and **??**.

(Step 4) Analysis of Sensitivity & Robustness. NormCoRe requires that translation choices introducing degrees of freedom be either controlled or systematically varied. Accordingly, the replication incorporates several robustness mechanisms. First, reproducibility controls (explicit random seeds, bounded discussion rounds, temperature parameters) enable deterministic reruns and isolate stochastic variation. Second, sensitivity analyses are embedded directly into the design: agent language (English, Mandarin, Spanish), is varied to assess the stability of normative outcomes. Third, structural constraints (e.g., validation of floor and range parameters, minimum statement lengths, and capped memory) prevent degenerate or nonsensical trajectories that would confound interpretation.

4.3 Results of the Replication Study

Difference Human group vs. AI agent group - Table 4: comparison humans vs. MAAI - Figure 9: preference ranking shifts

Table 1. Comparison of distributive justice principle choices between the baseline study of **?**] and MAAI experiments. AI groups demonstrate a higher consensus rate (90.9% vs. 79.4%) and stronger preference for the floor constraint principle (29 vs. 23 groups).

Principle	Baseline Study	MAAI
Max. floor income	1	0
Max. average income	1	1
Max. average with floor constraint	23	29
Max. average with range constraint	2	0
No Agreement	7	3
Total	**34**	**33**

Sensitivity: (1) Cognitive layer - Variance of foundation model/LLM choices (Table 5)

(2) Ontological layer - Variance of language in persona description (Table 9, Figure 14)

5 Discussion

The increasing integration of multiple AI agents in both workflow automation and scientific studies raises a range of novel challenges. NormCoRe provides a first step towards systemizing the integration of AI agents in replication studies to better understand the role of social and ethical norms in MAAI systems. Still, several open questions and challenges remain that future work should address.

5.1 Purpose of AI Agent Studies

- What value do we expect from replication studies with AI agents? Is it just for the sake of curiosity? Or are we drawing conclusions from that? - Can it contribute to the replication crisis? The scientific community currently faces a "Replication Crisis", characterized by the systematic failure to directly reproduce published findings as 90% of researchers across disciplines perceive their fields in such a crisis [5]. For example, Collaboration [11] replicated 100 psychological experiments, and failed in 54-64% (depending on the measurement) of the time with effect sizes averaging half the original magnitude. - Are we interested in social norms *among* AI agents or only in the outcomes related to humans? - Is it a good idea to conduct experiments with AI agents? -> there must be literature on this critique already - risks vs. benefits (e.g. sampling bias) - (MAAI) replication studies can methodologically improve upon baseline studies (e.g., because data can be collected more conveniently or some data becomes available that is not available for human subject studies)

- Implicit Hypothesis that AI Agents have a consistent "psychology" across domains - How they act in replication studies tells us something on how they act in applications e.g. if an AI Agent thinks that a woman has more empathy and social abilities in a psychological experiment that they also assign this property HR use case when assessing candidates CVs –> There is preliminary evidence that LLMs internally map concepts which support the hypothesis that LLMs process informations in concepts and supports the basic notion that what LLMs in one context can help us understand how the behave in different contexts [25]. That beeing said, reserach into this field is in an early stage.

5.2 Open Challenges of AI Agent Studies

- generalizability of findings - Schnelllebigkeit: Empirische Ergebnisse immer nur ein Snapshot der Geschichte - foundation models advance, human biology remains constant - establishing best practices - how should sample sizes be selected and reported in AI agent studies? - are there best practices for explicitation? -How detailed should researchers communicate their design decisions? Like on which level from general idea, to architectural plan to concrete code implementation? –> What is the "right" granularity?

- Are there (non-WEIRD) AI agents/MAAI systems that behave differently than our MAAI? - Are there (non-WEIRD) cultures in which AI agents/MAAI systems behave similarly to the human baseline? - Edge cases: hybrid replication in which we hold the human part constant and only translate the AI agent part - What about translation from MAAI to human-subject studies? - moral agents?

- What should be the goal of AI-agent replication studies: explanation, prediction, or alignment?
- How interpretable are human–AI differences given translation and explicitation choices?
- Do observed AI norms reflect model properties or experimental design artifacts?
- (How stable are normative findings across models, languages, and time?)
- Can NormCoRe meaningfully improve transparency and replicability in MAAI research?
- (What are the limits of treating AI agents as proxies for human groups?)
- When do AI group norms pose risks for real-world deployment?

6 Conclusion

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Appendix

Table 2. NormCoRe Translation Table: cognitive, onthological analogy

Aspect	Human Subject Study	Analogy Layer	Translation Choice	AI Agent Study Instantiation	Rationale
Participant Sampling	Convenience sample of university students	Cognitive	Explicitation (Incremental)	Selection of sufficiently capable foundation models and persona instantiations	Human sampling implicitly ensures sufficient cognitive and linguistic capacity to understand the task; in AI agent studies, this assumption must be made explicit by selecting foundation models with adequate reasoning, language comprehension, and instruction-following capabilities.
Statement Validation	Unconstrained natural speech	Cognitive	Explicitation (Incremental)	Minimum statement length with retry mechanism	Basic output constraints prevent degenerate or empty responses and improve robustness of the experimental artifact.
Participant Type	Human subjects (university students)	Ontological	Literal Translation (Direct)	LLM-based agents	Fundamental premise of the experiment: normative judgments are produced by deliberating agents embedded in groups.
Participant Personality	Natural human variation	Ontological	Explicitation (Incremental)	Configurable Role Description	Human personality variation is implicit in the original study; agents are explicitly instructed to behave like college students to approximate the participant pool.
Participant Memory	Human biological memory with natural limitations	Ontological	Explicitation (Constructive)	Agent-managed memory with character limits	Human memory is implicit and embodied; in AI agents it must be formalized to enable learning-in-context and consistency across deliberation rounds.
Participant Language	Primarly English (flawed Polish)	Ontological	Explicitation (Quasirandom)	Configurable prompt and agent language (English, Mandarin, Spanish)	Language is implicit and fixed in the human study; explicit variation enables testing sensitivity of normative outcomes to linguistic framing.
Temperature Control	*Not applicable*	Ontological	Explicitation (Comprehensive)	Configurable LLM temperature parameter for response randomness.	No human analogue exists; response stochasticity is explicitly parameterized to ensure reproducibility and controlled variation.

Table 3. NormCoRe Translation Table: interactional, interventional analogy

Aspect	Human Subject Study	Analogy Layer	Translation Choice	AI Agent Study Instantiation	Rationale
Experiment Duration	Unlimited Time	Inter-actional	Explicitation (Constructive)	Fixed maximum number of rounds (default: 10), with configurable upper bounds	LLM agents cannot speak simultaneously; therefore, the number of interaction rounds must be bounded to control computational cost.
Question Asking	Natural Questions to experimenter	Inter-actional	Explicitation (Incremental)	Not supported in dynamic orchestration	Supporting ad hoc questions would substantially increase system complexity, and it cannot be guaranteed that responding AI agents would provide accurate or authoritative information.
Discussion Format	Free-form group discussion	Inter-actional	Explicitation (Constructive)	Turn-based sequential discussion with equal speaking opportunities	Human conversational norms must be made explicit in AI systems; turn-based interaction ensures procedural fairness and comparability across agents independent of response latency.
Prompt Structure	Identity	Inter-ventional	Explicitation (Constructive)	At each interaction, AI agents receive an instruction prompt containing their name, role description, bank balance, current experimental phase, and memory state (in Phase 2 including the names of other participants and the discussion history), each separated by line breaks. The input prompt separately specifies the agent's current task.	High-level agent identity is explicitly separated from task-specific instructions to distinguish stable background characteristics from situational decision demands and to hierarchically structure information from global to local context.

Table 4. NormCoRe Translation Table: interventional analogy

Aspect	Human Subject Study	Analogy Layer	Translation Choice	AI Agent Study Instantiation	Rationale
Environment	Laboratory setting with physical presence	Interventional	Literal translation (Direct)	Computational / Data environment	The physical laboratory environment is directly translated into a computational setting, preserving the experimental context while adapting it to non-embodied agents.
Payment Mechanism	Real monetary payment based on chosen principle	Interventional	Explicitation (Constructive)	Symbolic payoff with explicit bank balance updates	Because AI agents cannot receive physical monetary payments, payoffs are explicitly represented symbolically via balance updates, introducing a new representational structure.
Probability Calculations	Original probability values	Interventional	Explicitation (Incremental)	Recalculated probabilities with explicit assumptions for Situation C	Minor assumptions are introduced to resolve underspecification in the original probability structure while preserving the intended payoff logic.
Distribution Presentation	Tabular presentation to subjects	Interventional	Literal translation (Direct)	Structured prompt-based presentation of distribution details	The informational content of the original tabular presentation is preserved and directly translated into a text-based prompt format.
Payoff Presentation	Single realized payoff and counterfactual outcomes	Interventional	Explicitation (Comprehensive)	Explicit presentation of realized and counterfactual payoffs for all principles	The payoff presentation is substantially expanded to fully explicate counterfactual information required for consistent interpretation by AI agents.
Floor / Range Constraint Options	Range of permissible floor and range constraint values	Interventional	Explicitation (Comprehensive)	Validated and constrained floor and range parameters with error handling	Implicit plausibility constraints in the human study are fully formalized through validation rules and corrective feedback.
Experimenter Instructions	Verbal instructions from human experimenter	Interventional	Explicitation (Constructive)	Structured JSON-based prompts across all experimental stages	Experimenter instructions are formalized into structured prompts to ensure consistency and reproducibility without introducing an artificial authority agent.
Distributions in Application Rounds	Four predefined situations (A–D)	Interventional	Explicitation (Incremental)	Original distributions adopted with minor probability assumptions	Some probability values are not explicitly specified in the original study and are inferred under minimal assumptions to render the distributions computationally executable.

Table 5. NormCoRe Translation Table: Interventional Analogy

Aspect	Human Subject Study	Analogy Layer	Translation Choice	AI Agent Study Instantiation	Rationale
Comprehension Test	Comprehension test to verify participant understanding	Interventional	Explicitation (Incremental)	Comprehension test omitted	Human comprehension checks ensure baseline task understanding; for AI agents, task comprehension is enforced through prompt design, making an explicit test redundant and unnecessarily complex.
Speaking Order	Emergent conversational order	Interventional	Explicitation (Incremental)	Randomized speaking order with constraints	Randomization mitigates systematic ordering effects (e.g., first- or last-mover advantages) while preserving the deliberative structure of the discussion.
Discussion History Management	Natural human memory recall	Interventional	Explicitation (Constructive)	Explicit provision of shared discussion history with bounded context length	Human memory is implicit and selective; in AI agents, shared memory must be explicitly provided and bounded to prevent context overflow while maintaining deliberative continuity.
Consensus Process	Verbal consensus and secret-ballot confirmation	Interventional	Explicitation (Constructive)	Formalized two-stage secret ballot with validation	Consensus formation is operationalized explicitly to ensure unambiguous termination and verifiable agreement in the absence of non-verbal cues.
Vote Initiation	Implicitly emerges during discussion	Interventional	Explicitation (Constructive)	Explicit per-round vote-initiation query with confirmation protocol	Human cues for vote initiation have no direct analogue; an explicit protocol ensures consistent triggering of the voting phase across models with varying capabilities.
Novel Principles	Participants could propose new principles	Interventional	Explicitation (Incremental)	Proposal of novel principles disabled	Allowing novel principles would substantially expand the decision space; this constraint mirrors empirical findings that no novel principles emerged in the original study.
Internal Thinking	Implicit cognitive deliberation	Interventional	Explicitation (Constructive)	Private strategic assessment prompt per round	Explicit internal reasoning prompts support structured deliberation by AI agents and compensate for the absence of implicit cognitive processes.
Payoff Calculation	Higher but unspecified stakes in group phase	Interventional	Explicitation (Incremental)	Original distributions scaled by random factor (2–6)	Because the original study specifies higher stakes without defining distributions, proportional scaling preserves the payoff logic while covering a plausible outcome range.